

Outward cusps refute fixed-exponent Sobolev self-improvement

Autonomous proof-discovery packet for arXiv:2604.23598

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Status. Candidate full counterexample, likely valid; urgent human review recommended. Model: GPT5.6.

1 Source theorem and question

Mishra and Mohanta define a bounded domain $\Omega \subset \mathbb{R}^n$ to be a $(1, s, p)$ -extension domain if there is a bounded extension

$$\dot{W}^{1,p}(\Omega) \longrightarrow \dot{W}^{s,p}(\mathbb{R}^n).$$

Their Theorem 1.6 claims that, when $p > 1$ and $1/p < s < 1$, such a domain is a $(1, p)$ -extension domain provided

$$\mathcal{H}^{n-1+(sp-1)/p}(\partial\Omega) = 0.$$

They then ask to what extent this Hausdorff-null hypothesis can be relaxed [1]. The exact passage from source PDF page 3 is reproduced below.

atically elegant, from a ‘practical’ perspective, very difficult to verify. So, we ask: what happens if the fractional parameter s is fixed? We use a stronger hypothesis than [Proposition 1.5](#) (for a fixed s) for this: $(1, s, p)$ -extension property.

Theorem 1.6. *Let $p \in (1, \infty)$ and $s \in (\frac{1}{p}, 1)$. Then every bounded $(1, s, p)$ -extension domain Ω , with $\mathcal{H}^{n-1+\frac{sp-1}{p}}(\partial\Omega) = 0$, is also a $(1, p)$ -extension domain.*

The lower bound of s is natural, as $W^{s,p}(\mathbb{R}^n)$, for $sp \leq 1$, is known to contain characteristic functions of certain domains. However, it is not clear to us, to what extent the assumption of $\mathcal{H}^{n-1+\frac{sp-1}{p}}$ measure of the boundary can be relaxed in the hypothesis of [Theorem 1.6](#).

We now discuss the paper’s organisation. [Section 2](#) contains necessary definitions, notations, conventions and some known results required for the subsequent sections. In [Section 3](#), we prove [Theorem 1.1](#) and [Theorem 1.3](#). In [Section 4](#), we prove [Theorem 1.6](#).

The result below is stronger than a sharpness answer: outward power cusps satisfy the displayed Hausdorff-null condition and still contradict the conclusion for every admissible parameter triple.

2 Proof intuition

First-order extension with a loss of integrability can be converted into fractional extension without that loss. More precisely, if

$$W^{1,p}(\Omega) \longrightarrow W^{1,q}(\mathbb{R}^n), \quad q > \frac{np}{n + (1-s)p},$$

then the strict Sobolev embedding $W^{1,q}(\mathbb{R}^n) \subset W^{s,p}(\mathbb{R}^n)$ finishes the conversion. Mild outward cusps admit the first operator for q arbitrarily close to p , but they fail the measure-density condition required of a first-order p -extension domain.

The source's boundary hypothesis does not exclude these cusps. Their boundary has finite $(n-1)$ -dimensional measure, so it has zero measure in the strictly larger dimension $n-1+s-1/p$. Thus the known integrability-loss extension theory supplies counterexamples inside, rather than just beyond, the claimed geometric regime.

3 A strict first-order-to-fractional embedding

We record the elementary version needed below.

Lemma 1. *Let $1 < q < p < \infty$, $0 < s < 1$, $q \neq n$, and*

$$\theta := 1 - \frac{n}{q} + \frac{n}{p} > s.$$

Then

$$[v]_{W^{s,p}(\mathbb{R}^n)} \leq C \|v\|_{W^{1,q}(\mathbb{R}^n)} \quad (v \in W^{1,q}(\mathbb{R}^n)).$$

Proof. Write $\Delta_h v = v(\cdot + h) - v$. If $q < n$, put $q^* = nq/(n-q)$. The inequality $\theta > 0$ says $q < p < q^*$. The translation estimate and Sobolev inequality give

$$\|\Delta_h v\|_q \leq |h| \|\nabla v\|_q, \quad \|\Delta_h v\|_{q^*} \leq 2 \|v\|_{q^*} \leq C \|\nabla v\|_q.$$

Interpolating to L^p gives

$$\|\Delta_h v\|_p \leq C |h|^\theta \|\nabla v\|_q \quad (|h| \leq 1),$$

because $1/p = \theta/q + (1-\theta)/q^*$. Also $\|v\|_p \leq C \|v\|_{W^{1,q}}$ by interpolation between L^q and L^{q^*} .

If $q > n$, Morrey's inequality and the L^q translation estimate give

$$\|\Delta_h v\|_\infty \leq C |h|^{1-n/q} \|\nabla v\|_q, \quad \|\Delta_h v\|_q \leq |h| \|\nabla v\|_q.$$

Interpolation between L^q and L^∞ again yields the same $|h|^\theta$ estimate, and $W^{1,q} \subset L^p$ follows similarly.

Finally use the difference representation

$$[v]_{W^{s,p}}^p = \int_{\mathbb{R}^n} \frac{\|\Delta_h v\|_p^p}{|h|^{n+sp}} dh.$$

For $|h| \leq 1$, the radial integral is a constant multiple of $\int_0^1 r^{(\theta-s)p-1} dr$, which is finite because $\theta > s$. For $|h| > 1$, use $\|\Delta_h v\|_p \leq 2 \|v\|_p$. $\square$

4 Counterexamples for every admissible parameter

Theorem 1. *Let $n \geq 2$, $1 < p < \infty$, and $1/p < s < 1$. There is a bounded domain $\Omega \subset \mathbb{R}^n$ such that*

1. Ω is a $(1, s, p)$ -extension domain;
2. $\mathcal{H}^{n-1+(sp-1)/p}(\partial\Omega) = 0$;
3. Ω is not a $(1, p)$ -extension domain.

Consequently Theorem 1.6 of [1], as stated in arXiv:2604.23598v1, is false, and its following relaxation question has no positive answer without changing some other hypothesis.

Proof. Choose $\tau > 1$ sufficiently close to one that

$$1 + (n-1)\tau < \min\{np, n + (1-s)p\}. \quad (1)$$

Then choose $1 < \beta < \tau$ and choose $q \neq n$ in the nonempty interval

$$\frac{np}{n + (1-s)p} < q < \frac{np}{1 + (n-1)\tau}. \quad (2)$$

The left endpoint in (2) is bigger than one because $sp > 1$ and $p > 1$, while the right endpoint is less than p . Hence $1 < q < p$.

Let $\psi(t) = t^\beta$ and take the capped outward cusp

$$\Omega = \{(t, x) \in (0, 1] \times \mathbb{R}^{n-1} : |x| < t^\beta\} \cup \{(t, x) \in [1, 2) \times \mathbb{R}^{n-1} : |x| < 1\}.$$

Koskela and Zhu's cusp-extension theorem [2, Theorem 1.2], reproduced in part in the supporting crop below, applies with its auxiliary exponent τ : indeed $\psi(t)/t = t^{\beta-1}$ is nondecreasing and tends to zero, and

$$\int_0^1 \left(\frac{t^\tau}{\psi(t)} \right)^{n/(\tau-1)} \frac{dt}{t} = \int_0^1 t^{n(\tau-\beta)/(\tau-1)-1} dt < \infty.$$

Moreover (1) gives $[1 + (n-1)\tau]/n < p$, and (2) gives $q < np/[1 + (n-1)\tau]$. Thus there is a bounded linear extension

$$E : W^{1,p}(\Omega) \longrightarrow W^{1,q}(\mathbb{R}^n). \quad (3)$$

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In [15, 16, 17], for certain Lipschitz cuspidal functions ψ , Maz'ya and Poborchii used integrability conditions on ψ to characterize the Sobolev extension property for Ω_ψ^n . By transferring an outward cuspidal domain onto a Lipschitz outward cuspidal domain via a global bi-Lipschitz transformation, we obtain a more general version of their result.

Theorem 1.2. *Let $\psi : (0, 1] \rightarrow (0, \infty)$ be a cuspidal function such that $\psi|_{(0,1]} > 0$ and the function $\psi(t)/t$ is nondecreasing on $(0, 1]$ with $\lim_{t \rightarrow 0} \psi(t)/t = 0$. We have following statements.*

(1) : *If*

$$(1.3) \quad \int_0^1 \left(\frac{t^s}{\psi(t)} \right)^{\frac{n}{s-1}} \frac{dt}{t} < \infty,$$

then there exists a bounded linear extension operator E_1 from $W^{1,p}(\Omega_\psi^n)$ to $W^{1,q}(\mathbb{R}^n)$ whenever $\frac{1+(n-1)s}{n} \leq p < \infty$ and $1 \leq q \leq \frac{np}{1+(n-1)s}$, and a bounded linear extension operator E_2 from $W^{1,p}(\Omega_\psi^n)$ to $W^{1,q}(\mathbb{R}^n)$ whenever $\frac{1+(n-1)s}{2+(n-2)s} \leq p < \infty$ and $1 \leq q \leq \frac{(1+(n-1)s)p}{1+(n-1)s+(s-1)p}$.

(2) : *If*

$$(1.4) \quad \int_0^1 \left(\frac{t^s}{\psi(t)} \right)^{\frac{n}{s-1}} \left| \log \left(\frac{\psi(t)}{t} \right) \right|^{-\alpha} \frac{dt}{t} < \infty$$

with $\alpha > (n-2)p$ then there exists a bounded linear extension operator E from $W^{1,p}(\Omega_\psi^n)$ to $W^{1,q}(\mathbb{R}^n)$.

Every outward cusp is a p -Poincaré domain; moreover, on such a cusp the nonhomogeneous $(W^{1,p}, W^{1,q})$ extension property is equivalent to the homogeneous $(L^{1,p}, L^{1,q})$ property [3, Theorem 1.2]. Concretely, for $u \in \dot{W}^{1,p}(\Omega)$ one may normalize by its average and set

$$Tu := E(u - u_\Omega) + u_\Omega.$$

The Poincaré inequality and (3) give

$$\|E(u - u_\Omega)\|_{W^{1,q}(\mathbb{R}^n)} \leq C \|u - u_\Omega\|_{W^{1,p}(\Omega)} \leq C \|\nabla u\|_{L^p(\Omega)}.$$

For $v = E(u - u_\Omega)$, condition (2) is exactly

$$1 - \frac{n}{q} + \frac{n}{p} > s.$$

Lemma 1 therefore gives

$$[Tu]_{W^{s,p}(\mathbb{R}^n)} = [v]_{W^{s,p}(\mathbb{R}^n)} \leq C \|\nabla u\|_{L^p(\Omega)}.$$

Since $Tu = u$ on Ω , this proves that Ω is a $(1, s, p)$ -extension domain.

The lateral boundary has finite $(n-1)$ -dimensional measure: its area is bounded by a constant times

$$\int_0^1 t^{\beta(n-2)} \sqrt{1 + \beta^2 t^{2\beta-2}} dt < \infty,$$

and the capped part is Lipschitz. Hence $\mathcal{H}^{n-1}(\partial\Omega) < \infty$. Since $(sp-1)/p = s-1/p > 0$, this implies

$$\mathcal{H}^{n-1+(sp-1)/p}(\partial\Omega) = 0.$$

It remains to exclude first-order p extension. For small $r > 0$, the point $z_r = (r, 0)$ belongs to Ω , and

$$|B(z_r, r) \cap \Omega| \leq C \int_0^{2r} t^{\beta(n-1)} dt \leq Cr^{1+\beta(n-1)} = o(r^n),$$

because $\beta > 1$. Thus Ω fails the measure-density condition. Every bounded $(1, p)$ -extension domain satisfies that condition [4, Theorem 6]; hence Ω is not a $(1, p)$ -extension domain. $\square$

5 A fully numerical planar instance

Take

$$n = p = 2, \quad s = \frac{3}{4}, \quad \tau = \frac{5}{4}, \quad \beta = \frac{6}{5}, \quad q = \frac{5}{3}.$$

Then

$$\int_0^1 \left(\frac{t^{5/4}}{t^{6/5}} \right)^{2/(5/4-1)} \frac{dt}{t} = \int_0^1 t^{-3/5} dt = \frac{5}{2},$$

and

$$\frac{1+\tau}{2} = \frac{9}{8} < 2, \quad q = \frac{5}{3} < \frac{4}{1+\tau} = \frac{16}{9}.$$

Thus $W^{1,2}(\Omega) \rightarrow W^{1,5/3}(\mathbb{R}^2)$. The translation exponent in Lemma 1 is

$$1 - \frac{2}{5/3} + \frac{2}{2} = \frac{4}{5} > \frac{3}{4},$$

so Ω is a $(1, 3/4, 2)$ -extension domain. Its boundary has finite length, and therefore $\mathcal{H}^{5/4}(\partial\Omega) = 0$, but its density at the cusp tip decays like $r^{11/5}/r^2 = r^{1/5}$.

6 Scope, novelty, and review recommendation

This packet challenges a theorem in a very recent preprint, so the novelty claim is necessarily provisional. The four run indexes and packet/attempts trees were searched for arXiv:2604.23598, fixed-exponent fractional extension, and outward cusps. Bounded arXiv/web searches through 13 August 2026 used the exact source id and title, “Theorem 1.6”, “correction”, “error”, “counterexample”, “ $(1, s, p)$ -extension”, and “outward cusp”. They found the source and the background papers used here, but no correction or prior statement of this counterexample.

No numerical computation is used. Expert review should first check the exact parameter substitution into [2, Theorem 1.2], the homogeneous normalization justified by [3, Theorem 1.2], and the translation proof of Lemma 1. If these pass, the source’s proof of Theorem 1.6 must contain a substantive failure; locating that failure is logically separate from the counterexample.

References

- [1] R. Mishra and K. Mohanta, *How to Recognise Extension domains*, arXiv:2604.23598v1, 2026.
- [2] P. Koskela and Z. Zhu, *The extension property for domains with one singular point*, arXiv:2110.07565v1, 2021.
- [3] P. Koskela, R. Mishra, and Z. Zhu, *Sobolev versus homogeneous Sobolev extension II*, arXiv:2507.07072v2, 2025.
- [4] P. Hajłasz, P. Koskela, and H. Tuominen, *Measure density and extendability of Sobolev functions*, Rev. Mat. Iberoam. **24** (2008), 645–669.